IEEE Access
Multidisciplinary ⁝ Rapid Review ⁝ Open Access Journal

Received 20 January 2025, accepted 4 February 2025, date of publication 5 February 2025, date of current version 13 February 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3539502

RESEARCH ARTICLE

Fine-Tuning EEG Channel Utilization for Emotionally Stimulated Biometric Authentication

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This work was supported by the Life Sciences Research Board (LSRB), Defence Research and Development Organisation under Grant LSRB/01/15001/LSRB-401/LS&BD/2023.

ABSTRACT Biometric authentication relies on distinct biological traits of individuals to validate their identity, enhancing security measures. However, variations in an individual's emotional state can impact the reliability of the biometric system. In this study, we propose a novel pipeline to evaluate electroencephalography (EEG)-based biometric system across different emotional states and optimize critical brain regions using machine learning algorithms. EEG signals from the DEAP dataset were classified into four emotional states: HAHV, HALV, LALV, and LAHV. We extracted a comprehensive set of statistical, time, frequency, entropy, fractal, spectral, and shape features from each channel. Machine learning classifiers, including Random Forest, Gradient Boosting, Extreme Gradient Boosting, LightGBM, CatBoost, and Bagging, were used for participant authentication. Our results revealed that the CatBoost classifier performed well across all stimuli with average accuracies of 84%, 85%, 86%, and 83% for HAHV, HALV, LALV, and LAHV, respectively. We found that features from channels FC1, Fz, C4 & Pz, and FC1 significantly contributed to EEG authentication on stimuli such as HAHV, HALV, LALV, and LAHV, respectively. Features such as skewness and the theta-to-alpha frequency band ratio consistently performed well across stimuli, demonstrating EEG signals' potential for robust biometric authentication by addressing emotional variations.

INDEX TERMS Biometric, electroencephalography, emotions, brain location, machine learning.

I. INTRODUCTION

Biometric authentication encompasses various modalities that utilize distinctive biological traits for identity verification. There are primarily two categories of biometric systems: conventional and cognitive biometrics. Conventional biometrics use only physiological characteristics, which include fingerprints [1], iris patterns [2], facial characteristics [3], and voice recognition [4] for authentication. These methods, however, have their drawbacks, such as susceptibility to replication or forgery and potential issues with accuracy under various physical conditions or with changes over time [5].

The associate editor coordinating the review of this manuscript and approving it for publication was Vincenzo Conti.

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Neural Computing and Applications (2024) 36:12565–12585
https://doi.org/10.1007/s00521-024-09770-3

ORIGINAL ARTICLE

Autism spectrum disorder diagnosis using fractal and non-fractal-based functional connectivity analysis and machine learning methods

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Received: 21 November 2023 / Accepted: 25 March 2024 / Published online: 22 April 2024
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Abstract

Autism spectrum disorder (ASD) is a neurological condition characterized by impaired functional connectivity (FC) networks in the brain. There are several brain networks associated with ASD that have been studied for ASD diagnosis, but the results are inconsistent. A functional magnetic resonance imaging (fMRI) study was performed to address this gap by comparing brain networks among autistic individuals and individuals with typical development (TD) using data from the ABIDE-I and ABIDE-II databases. Blood oxygen level-dependent (BOLD) time series were extracted from 236 regions of interest (ROI) in fMRI data using three atlases: Gordon's, Harvard Oxford, and Diedrichsen. Consequently, 27,730 nonlinear features are extracted from FC matrices, including fractals, non-fractals, and Pearson correlation coefficients (PCC). A parametric and nonparametric classifier was used to analyze the top 0.1%, 0.3%, 0.5%, 0.7%, 1%, 2%, and 3% of features based on the XGBoost feature ranking algorithm. In the study, we found that non-fractal brain FC measures can accurately identify ASD and TD more effectively than fractal and PCC measures. Classifiers performed well, with FC features at the top 0.3%. The classification model at OHSU was more accurate than the model at other sites. There was 100% accuracy at a single site and 96.17% accuracy at all sites using a multilayer perceptron classifier with non-fractal features. The classifier model shows that Cingulo-Parietal Task Control (13.6%), Retro-Splenial Temporal (RST) (13.3%), and Salience (11.6%) are significant contributors. The optimal performance was observed in features derived from networks such as RST and default (4 connections), auditory and Fronto-Parietal Task Control (3 connections), Cingulo-Opercular Task Control (COTC), and ventral attention (3 connections) within COTC (3 connections), visual and COTC (3 connections), and cerebellum and COTC (3 connections). Based on the results, non-fractal-based FC has value in distinguishing ASD from TD using resting-state fMRI.

Keywords Autism spectrum disorder · Resting-state fMRI · Functional connectivity · Fractal · Non-fractal · Machine learning

1 Introduction

The autism spectrum disorder (ASD) is a neurodevelopmental disorder characterized by restricted or repetitive behaviors, as well as atypical social interactions and communication [1]. Across the globe, ASD prevalence varies from region to region, with a median rate of 100 cases per 10,000 people [2]. ASD prevalence varies geographically, with 0.4%, 1%, 0.5%, 1%, and 1.7%, respectively, in Asia, America, Europe, Africa, and Australia [3]. Various studies have suggested that ASD results from the interaction between genes and the environment [4]. Diagnostic criteria for autism include clinical scores and behavioral symptoms; however, behavioral assessments require a highly experienced and subjective multidisciplinary team, which can take weeks or months. Furthermore, it could result in poor treatment due to a lack of understanding of neuropathology [5]. It is important to differentiate typically developing (TD) individuals from those with ASD so that early diagnosis and intervention can improve ASD individuals' outcomes. In addition, researchers may be able to identify neurological distinctions between individuals with TD and individuals with ASD, potentially leading to more effective interventions. It is critical to establish an early and accurate diagnosis

Extended author information available on the last page of the article

Arabian Journal for Science and Engineering (2024) 49:6847–6865
https://doi.org/10.1007/s13369-023-08560-8

RESEARCH ARTICLE-ELECTRICAL ENGINEERING

Age- and Severity-Specific Deep Learning Models for Autism Spectrum Disorder Classification Using Functional Connectivity Measures

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Received: 10 August 2023 / Accepted: 22 November 2023 / Published online: 12 December 2023
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Abstract

Autism spectrum disorder (ASD) is characterized by divergent etiological factors, comorbidities, severity levels, genetic influences, and functional connectivity (FC) patterns in the brain. In the literature, ASD classification based on age and severity using fMRI data is extremely limited. This study explores the impact of age, symptom severity, and brain FC patterns on the diagnosis of ASD using deep neural networks (DNNs). The ability to classify ASD by age and severity using fMRI data is extremely limited. This study explores the impact of age, symptom severity, and brain FC patterns on the diagnosis of ASD using deep neural networks (DNNs). The FC measures were extracted using Pearson's correlation coefficient (PCC), fractal connectivity (FrC), and nonfractal connectivity (NFrC) from the ABIDE I and II databases. We studied three age groups (6 to 11, 11 to 18, and 6 to 18 years) and two severity groups (ADOS score $\leq$ 11 and ADOS score > 11). The FC matrices are constructed from blood oxygen level-dependent (BOLD) time series signals, and the heat maps are used to generate features for the convolutional neural network (CNN), MobileNetV2, and DenseNet201 models. The MobileNetV2 classifier achieved 76.25% accuracy, 77.09% sensitivity, and 79.77% precision in the age group of 6 to 11 years using NFrC feature maps compared to other DNNs. According to ADOS total scores above 11, DenseNet201 demonstrated superior performance with 83.45% accuracy, 87.3% sensitivity, and 79.13% precision. Connectivity measured by NFrC consistently outperformed Frc measures. Various combinations of connectivity measures and classifiers consistently showed promising results for the age group of 6–11 years and the severity group with an ADOS score of more than 11. ASD's inherent heterogeneity can be addressed effectively by developing diagnostic models tailored to age and severity.

Keywords Autism spectrum disorder · Age and severity · Fractal and non-fractal connectivity · Deep learning

1 Introduction

As a neurodevelopmental disorder, autism spectrum disorder is characterized by impairments in social communication and interaction, as well as constrained, repetitive patterns of behavior, interests, or activities [1, 2]. There has been an increase in its prevalence over the last decade [3], with projections ranging from one in 59 to one in 54 children in the USA [4]. Despite its high prevalence, diagnosing ASD remains challenging. There are a variety of diagnostic tools available today, including psychological assessments, medical evaluations, and caregiver interviews, which are highly subjective and susceptible to misdiagnosis or overdiagnosis. In individuals with autism spectrum disorder (ASD), it is usually associated with local underconnectivity or overconnectivity in the cortex. In this sense, the automated identification of sig-

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IJARSCT
ISSN (Online) 2581-9429
International Journal of Advanced Research in Science, Communication and Technology (IJARSCT)
Impact Factor: 4.819
Volume 6, Issue 1, June 2021

Calophyllum Oil Prospective as Alternate Fuel for Diesel Engine

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Abstract: *Rapid depletion of fossil fuels, increasing fossil-fuel price, carbon price, and the quest of low carbon fuel for cleaner environment – these are the reason researchers are looking for alternatives of fossil fuels. Biodiesel is a gifted substitute as an alternative fuel has gained significant attention due to the predicted littleness of conventional fuels and environmental concern. The utilization of liquid fuels such as biodiesel produced from Calophyllum inophyllum oil by transesterification process represents one of the most promising options for the use of conventional fossil fuels. The Calophyllum inophyllum oil is converted into Calophyllum inophyllum methyl ester known as biodiesel processed in the presence of homogeneous acid catalyst. The physical properties such as Kinematic viscosity, Density, Calorific Value, Cetane number, Fire point and Flash point were found out for Calophyllum inophyllum methyl ester at different blends.*

Keywords: Calophyllum inophyllum, transesterification, Biodiesel, Alternative fuel, Emission

I. Introduction

In 2040, the demand for global energy will be about 30% higher than that of in 2010. In 2020, the global demand for crude oil is estimated to be 101.6 million barrels per day. The fossil fuels account for about 86% of the primary energy need which is increasingly rapidly. Most industries depend upon diesel as fuel for their energy needs. Mankind relies heavily on fuels for every aspect of their everyday life, good transport and electricity in particular. Since the last few decades, there is a drastic increase in demand of fuel, price and no. of a road vehicle, climate changes due to air pollutions and now it has become a serious topic for the researchers to guarantee the availability of energy alternatives and reduction in environmental pollution due to vehicular emissions.

1.1 Need of Biodiesel

Biodiesels are gaining worldwide acceptance because of the faster depleting petro-diesel fuels. Biodiesels are biodegradable, renewable and more environment friendly than petroleum based fuels [1].

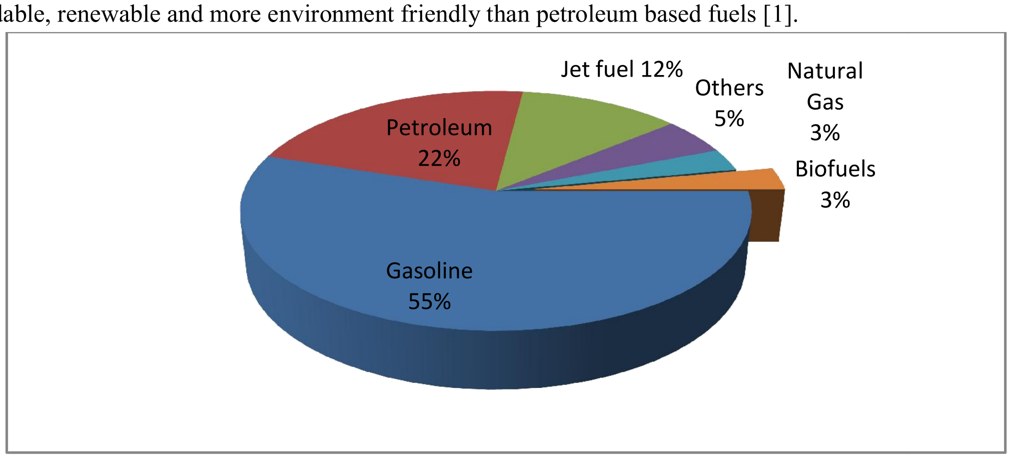

Figure 1: Transportation Energy Consumption

Healthcare Transformation with Informatics and Artificial Intelligence
J. Mantas et al. (Eds.)
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doi:10.3233/SHTI230424

Diagnostic Classification of ASD Using Fractal Functional Connectivity of fMRI and Logistic Regression

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Abstract. Our study used functional magnetic resonance imaging and fractal functional connectivity (FC) methods to analyze the brain networks of Autism Spectrum Disorder (ASD) and typically developing participants using data available on ABIDE databases. Blood-Oxygen-Level-Dependent time series were extracted from 236 regions of interest of cortical, subcortical, and cerebellar regions using Gordon's, Harvard Oxford, and Diedrichsen atlases respectively. We computed the fractal FC matrices which resulted in 27,730 features, ranked using XGBoost feature ranking. Logistic regression classifiers were used to analyze the performance of the top 0.1%, 0.3%, 0.5%, 0.7%, 1%, 2%, and 3% of FC metrics. Results showed that 0.5% percentile features performed better, with average 5-fold accuracy of 94%. The study identified significant contributions from dorsal attention (14.75%), cingulo-opercular task control (14.39%), and visual networks (12.59%). This study could be used as an essential brain FC method to diagnose ASD.

Keywords. Autism spectrum disorder, functional magnetic resonance imaging, fractal functional connectivity, logistic regression

1. Introduction

Autism spectrum disorder (ASD) is a neurodevelopmental disorder that affects social interaction and communication. It is characterized by delayed and disordered language, limited and stereotypical patterns of behavior, interests, and activities, and onset before the age of three. The complexities of ASD make the diagnosis challenging, and identifying a biomarker linked to ASD is crucial for early detection and targeted treatment. Several neuroimaging studies have used non-invasive brain imaging modalities, including structural magnetic resonance imaging (sMRI), electroencephalogram (EEG), magnetoencephalography, and diffusion tensor imaging, but functional magnetic resonance imaging (fMRI) [1] is one of the most promising modalities.

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Healthcare Transformation with Informatics and Artificial Intelligence
J. Mantas et al. (Eds.)
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doi:10.3233/SHTI230425

Diagnostic Classification of ASD Improves with Structural Connectivity of DTI and Logistic Regression

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Abstract. In this study, we examined the structural connectivity (SC) of autism spectrum disorder (ASD) and typical development using the distance correlation and machine learning algorithm. We preprocessed diffusion tensor images using a standard pipeline and parcellated the brain into 48 regions using atlas. We derived diffusion measures in white matter tracts, such as fractional anisotropy, radial diffusivity, axial diffusivity, mean diffusivity, and mode of anisotropy. Additionally, SC is determined by the Euclidean distance between these features. The SC were ranked using XGBoost and significant features were fed as the input to the logistic regression classifier. We obtained an average 10-fold cross-validation classification accuracy of 81% for the top 20 features. The SC computed from the anterior limb of internal capsule L to superior corona radiata R regions significantly contributed to the classification models. Our study shows the potential utility of adopting SC changes as the biomarker for the diagnosis of ASD.

Keywords. Autism spectrum disorder, diffusion tensor imaging, structural connectivity,logistic regression

1. Introduction

Autism spectrum disorder (ASD) is a neurodevelopmental disorder characterized by impaired communication, reciprocal social interaction, restricted and stereotyped pattern of behavior. The presence of these impairments is variable in range and severity and most often changes with the acquisition of other developmental diseases. ASD is diagnosed based on subjective clinical scores, which can be time-consuming and challenging, with limited accuracy. Studies have demonstrated that structral connectivity (SC) modeled by tracking the neural fibers between the brain regions could be valuable tool for diagnosing

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2023 IEEE 5th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA) | 979-8-3503-3828-7/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICCCMLA58983.2023.10346926

Advancing ASD Diagnostic Classification with Features of Continuous Wavelet Transform of fMRIand Machine Learning Algorithms

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***Abstract*— In this study, we aimed to develop a diagnostic classification model for autism spectrum disorder (ASD) using scalograms of functional magnetic resonance imaging (fMRI) data. Initially, the fMRI data from the ABIDE-I and ABIDE-II databases were preprocessed, and brain regions of interest (ROIs) were parcellated. Subsequently, scalograms were generated using Continuous Wavelet Transform from the average time series of each ROI. Various features were extracted from these scalogram images, including gray-level co-occurrence matrix, gray-level run-length matrix, fractal dimension texture analysis, Zernike's moments, Hu's moments, and first-order statistics. We employed CatBoost and Decision Tree machine learning classifiers and performed extensive five-fold cross-validation with hyperparameter tuning to evaluate the model's performance. The results indicated an overall accuracy of 83.04%, sensitivity of 66.66%, specificity of 85.57%, and an F1-score of 75.85% for the Decision Tree classifier, particularly a region in the Retro-Splenial Temporal network. In addition, regions from networks such as fronto-parietal task control network, Ventral Attention, and Visual, showcased promising classification performance for ASD. These findings highlight the efficacy of the proposed model in providing accurate ASD classification, which holds significant implications for early diagnosis and intervention strategies.**

Keywords*— *autism spectrum disorder, fMRI, scalograms, feature extraction, machine learning

I. Introduction

Autism spectrum disorder (ASD) is a lifelong neurodevelopmental condition that emerges during childhood and is characterized by three primary features: difficulties in forming social relationships, challenges in communication, and repetitive behavioral patterns. These deficits are believed to be linked to impaired executive functions, including working memory, inhibition, mental flexibility, and planning. ASD affects approximately 1 in 100 children worldwide, and due to its diverse nature, personalized therapies are necessary since there is no cure [1]. Detecting and treating ASD early are crucial steps in reducing symptoms and improving the quality of life for affected individuals. However, currently, there are no medical tests available for the detection of autism. Instead, ASD symptoms are typically identified through observation, with parents, teachers, and medical professionals such as pediatricians, child psychologists, neurologists, and therapists involved in the evaluation process. In adults, identifying ASD symptoms can be more challenging than in older children and adolescents, as some symptoms may overlap with other mental health disorders. Despite behavioral changes in children as early as 6 months of age, the diagnostic process remains complicated due to the wide range of behavioral patterns and subjective symptoms associated with ASD, leading to delays in diagnosis that can extend from months to years. Consequently, there is a need to explore autism-specific brain imaging biomarkers to facilitate early detection of ASD [2].

Magnetic resonance examination is a valuable tool for investigating brain structure changes in children with ASD. Functional brain imaging, particularly functional magnetic resonance imaging (fMRI), has been extensively utilized to study overall brain functioning. These advanced neuroimaging techniques can detect subtle alterations in brain metabolism and neural network connectivity in specific regions over time. These changes can be induced by cognitive tasks or can be observed during the brain's spontaneous activity at rest. Blood-oxygen-dependent level (BOLD) fMRI, which measures changes in deoxyhemoglobin concentration, allows visualization of these neural metabolism changes related to cognitive tasks or spontaneous brain activity [3]. In this study, we utilized resting state-fMRI data to identify specific brain biomarkers associated with ASD.

Functional Connectivity (FC) is a widely used technique to compute the statistical relationship between BOLD signals in different brain regions. However, it's important to acknowledge that the BOLD signal is noisy and only partially reflects neural activity. It arises from various physiological processes like neuronal, metabolic, cardiac, and vigilance functions, and can be affected by machine and participant-specific characteristics. In FC analyses, these confounding effects can be particularly troublesome because they can introduce biased FC estimates due to high correlations between noise sources in different brain regions [3]. Resting-state activity in fMRI also shows nonstationary BOLD signal fluctuations, which create challenges when using traditional methods like Discrete Fourier transform (DFT) or short-time Fourier transform (STFT) for time-frequency (TF) analysis. DFT requires stationarity of the brain signal, which is not always the case, and STFT can be inconvenient due to the number of samples and sample frequency required [4]. continuous Wavelet Transform (CWT) allows for optimal

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